

An Exact Audit of an Announced Three-Dimensional Keller Map

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Abstract

We independently audit the polynomial map $F = (P, Q, R) : \mathbb{A}^3 \rightarrow \mathbb{A}^3$ announced by Levent Alpoge on 20 July 2026. Exact calculation gives $\det DF = -2$, while three distinct rational points map to $(-1/4, 0, 0)$. Thus the map is a counterexample to the Jacobian Conjecture in dimension three. We derive two inverse cubics, prove generic degree three and full S_3 monodromy, classify the complex fibers, determine the image and nonproperness set, describe a finite smooth completion, and construct direct nonautomorphic endomorphisms of the third Weyl and Poisson algebras. We also exhibit a weighted scaling symmetry and collision family, a commuting divergence-free polynomial frame with no complete constant direction, a nonproper smooth fibration with jumping component count, an equivariant quotient square law, exact finite-field fiber statistics, and exact real volume formulas. We separate proved consequences from open directions and priority-sensitive claims. Exact symbolic scripts accompany this note. This is an independent, non-peer-reviewed audit and makes no claim of discovery or priority.

1 Source, status, and the exact counterexample

The primary public source located by our search is Alpoge’s announcement [1]. The post credits Akhil for the question and Fable for work on the example. Several structural observations also appeared in same-day public notes [14, 15, 16]; we therefore make no priority claim. As of the date above we did not locate a peer-reviewed paper or an arXiv manuscript by the announcer.

Define

$$\begin{aligned} P &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

Theorem 1.1. *The map $F = (P, Q, R)$ satisfies $\det DF = -2$ and*

$$F(0, 0, -1/4) = F(1, -3/2, 13/2) = F(-1, 3/2, 13/2) = (-1/4, 0, 0).$$

Its component degrees are 7, 6, 4.

Proof. Differentiate and expand the determinant, then substitute the three points. These are identities in $\mathbb{Z}[x, y, z, 1/2]$. The accompanying verifier performs the calculation using exact symbolic arithmetic. $\square$

Corollary 1.2. *The Jacobian Conjecture is false in dimension three and in every dimension $n \geq 3$. Replacing P by $-P/2$ normalizes the determinant to one. The result holds over every characteristic-zero field. It does not decide the two-dimensional case.*

1.1 Weighted equivariance and a collision family

For $\lambda \in \mathbb{C}^*$, define

$$\sigma_\lambda(x, y, z) = \left(\lambda x, \frac{y}{\lambda}, \frac{z}{\lambda^2} \right), \quad \tau_\lambda(p, q, r) = \left(\frac{p}{\lambda^2}, \frac{q}{\lambda}, \lambda r \right).$$

Exact substitution gives $F \circ \sigma_\lambda = \tau_\lambda \circ F$. Hence for every nonzero λ ,

$$\begin{aligned} F\left(0, 0, -\frac{1}{4\lambda^2}\right) &= F\left(\lambda, -\frac{3}{2\lambda}, \frac{13}{2\lambda^2}\right) \\ &= F\left(-\lambda, \frac{3}{2\lambda}, \frac{13}{2\lambda^2}\right) = \left(-\frac{1}{4\lambda^2}, 0, 0\right). \end{aligned}$$

The discriminant below is equivariant: $\Delta(\tau_\lambda(p, q, r)) = \lambda^{-2}\Delta(p, q, r)$. Since $\Delta(-1/4, 0, 0) = 4$, the announced collision is a regular generic three-sheet fiber, not an exceptional discriminant fiber. The symmetry $F(-x, -y, z) = (P, -Q, -R)$ is the case $\lambda = -1$.

1.2 The two-dimensional weighted quotient

The source and target invariant rings are

$$\mathbb{C}[x, y, z]^{\mathbb{G}^m} = \mathbb{C}[u, v], \quad u = xy, \quad v = x^2z, \quad \mathbb{C}[p, q, r]^{\mathbb{G}^m} = \mathbb{C}[\alpha, \beta], \quad \alpha = pr^2, \quad \beta = qr.$$

Every invariant monomial factors through these generators. Equivariance thus induces $\bar{F} : \mathbb{A}_{u,v}^2 \rightarrow \mathbb{A}_{\alpha,\beta}^2$. Put $h = 2 - 3u - v$. Exact elimination gives

$$\begin{aligned} \alpha &= (1 + u)h^2(3u^3 + u^2v + 4u^2 + 2uv + v), \\ \beta &= h(9u^3 + 3u^2v + 12u^2 + 6uv + u + 3v), \end{aligned}$$

and

$$\det D\bar{F} = -2h^2.$$

The distinct points $(0, 0)$ and $(-3/2, 13/2)$ both map to the quotient origin; indeed the whole critical line $h = 0$ maps there. Thus the quotient retains a shadow of noninjectivity but introduces ramification. It is not a plane Keller counterexample, and the formula pinpoints why this natural reduction fails.

Theorem 1.3 (Equivariant quotient square law). *Let $G = (A, B, C) : \mathbb{A}^3 \rightarrow \mathbb{A}^3$ be equivariant for source weights $(1, -1, -2)$ and target weights $(-2, -1, 1)$, and suppose $\det DG = c \in \mathbb{C}^*$. With $u = xy$, $v = x^2z$, $\alpha = AC^2$, and $\beta = BC$, write $C = xh(u, v)$. Then the induced quotient map satisfies*

$$\det D\bar{G} = ch(u, v)^2.$$

Thus a nonconstant C/x forces a critical divisor in the quotient.

Proof. For the infinitesimal orbit fields $\xi = (x, -y, -2z)$ and $\eta = (-2A, -B, C)$, direct exterior algebra gives

$$du \wedge dv = x^2 \iota_\xi(dx \wedge dy \wedge dz), \quad d\alpha \wedge d\beta = C^2 \iota_\eta(dA \wedge dB \wedge dC).$$

Equivariance gives $DG(\xi) = \eta \circ G$, and the constant Jacobian pulls the target volume form back to c times the source volume form. Pullback of the second identity and comparison with the first gives $\det D\bar{G} = c(C/x)^2 = ch^2$. $\square$

This is a theorem for the displayed weight pattern, not for arbitrary quotients or every possible descent to dimension two.

2 Inverse cubics and generic degree

Write target coordinates as (p, q, r) . In the function field, set $t = y + 1/x$. Direct simplification gives

$$q = 4t + \frac{2}{x} - 3rt^2, \quad 2p = rt^3 - 2t^2 + qt.$$

Thus t is a root of

$$C_{p,q,r}(T) = rT^3 - 2T^2 + qT - 2p. \quad (1)$$

If $\rho = C'(t)$, then

$$\rho = \frac{2}{x}, \quad x = \frac{2}{\rho}, \quad y = t - \frac{\rho}{2}, \quad z = \frac{5}{4}\rho^2 - \frac{3}{2}t\rho - \frac{r}{8}\rho^3. \quad (2)$$

Consequently $\mathbb{C}(x, y, z) = \mathbb{C}(p, q, r)(t)$ and the field degree is at most three. The three simple points in Theorem 1.1 have disjoint analytic neighborhoods mapped biholomorphically onto a common neighborhood. Hence the generic degree is at least three, and therefore exactly three. In particular, (1) is irreducible over $\mathbb{C}(p, q, r)$.

A projectively complete form is obtained from $s = x/(1 + xy)$:

$$H_{p,q,r}(s) = 2ps^3 - qs^2 + 2s - r = 0, \quad (3)$$

with homogenization

$$\Phi_{p,q,r}(S, T) = 2pS^3 - qS^2T + 2ST^2 - rT^3.$$

For a finite simple root s , put $d = 1 - sq + 3ps^2 = H'(s)/2$. Then

$$x = \frac{s}{d}, \quad y = q - 3ps, \quad z = \frac{2x - 3x^2y - r}{x^3} \quad (s \neq 0).$$

For $s = 0$, necessarily $r = 0$, and the regular formula is $(x, y, z) = (0, q, p - 4q^2)$. For the projective root $[S : T] = [1 : 0]$, simplicity forces $p = 0$ and $q \neq 0$, and

$$x = \frac{2}{q}, \quad y = -\frac{q}{2}, \quad z = \frac{5x - r}{x^3}.$$

Direct substitution checks both exceptional formulas. Conversely an affine point with $1 + xy \neq 0$ gives $H'(s) = 2/(1 + xy) \neq 0$, while a point with $1 + xy = 0$ gives a simple root at infinity. Thus simple projective roots are in bijection with affine preimages.

3 Discriminant, monodromy, fibers, and image

The discriminant of (1) is $-4\mathcal{Q}$, where

$$\mathcal{Q} = 27p^2r^2 - 18pqr + 16p + q^3r - q^2, \quad \text{Disc}_p(\mathcal{Q}) = -4(3qr - 4)^3.$$

Gauss's lemma shows that $\mathcal{Q}$ is irreducible and not a square in $\mathbb{C}(p, q, r)$. Since the cubic is irreducible,

$$\text{Gal}(C/\mathbb{C}(p, q, r)) \cong S_3.$$

The degree-three cover therefore has full S_3 monodromy. Its rational deck group is trivial because for the nonnormal cubic subextension L/K , $\text{Aut}_K(L) = N_{S_3}(S_2)/S_2 = 1$.

For (3), write

$$\Delta = q^2 - 16p - q^3r + 18pqr - 27p^2r^2.$$

Its discriminant is 4Δ . A cubic has no simple projective root precisely when it has a triple root. Coefficient comparison gives the triple-root curve

$$\Gamma = \left\{ \left(\frac{1}{3u^2}, \frac{2}{u}, \frac{2u}{3} \right) : u \in \mathbb{C}^* \right\} = V(12p - q^2, 3qr - 4).$$

Theorem 3.1. *For $(p, q, r) \in \mathbb{C}^3$,*

$$\#F^{-1}(p, q, r) = \begin{cases} 3, & \Delta \neq 0, \\ 1, & \Delta = 0 \text{ and } (p, q, r) \notin \Gamma, \\ 0, & (p, q, r) \in \Gamma. \end{cases}$$

Hence $F(\mathbb{C}^3) = \mathbb{C}^3 \setminus \Gamma$.

Proof. Off $V(\Delta)$ the projective cubic has three simple roots. On the discriminant but off Γ , it has a double root and a simple root. The simple root reconstructs one affine point; the double root cannot, because F is etale and has no ramified affine point. On Γ the unique root is triple, so no affine preimage exists. $\square$

Generic degree three is minimal for any Keller counterexample. Degree one is the birational case; a separable degree-two extension is Galois, and the Galois case of the Jacobian Conjecture is known [3, 4, 5].

4 Infinity and the smooth finite completion

The explicit path

$$(x, y, z) = \left(\lambda, -\frac{1}{\lambda}, \frac{5}{\lambda^2} \right), \quad F(x, y, z) = \left(0, \frac{2}{\lambda}, 0 \right)$$

proves nonproperness.

Define

$$I = \{((p, q, r), [S : T]) \in \mathbb{A}^3 \times \mathbb{P}^1 : \Phi_{p,q,r}(S, T) = 0\}.$$

The projection $\pi : I \rightarrow \mathbb{A}^3$ is finite of degree three. The partial derivatives with respect to p, q, r are $2S^3, -S^2T, -T^3$, which never vanish simultaneously, so I is smooth. The map

$$(x, y, z) \mapsto (F(x, y, z), [x : 1 + xy])$$

identifies $\mathbb{A}^3$ with $I \setminus D$, where D is the multiple-root divisor.

On the chart $S = 1$, solving $\Phi = \partial\Phi/\partial u = 0$ gives

$$p = u^2 - ru^3, \quad q = 4u - 3ru^2.$$

Thus $D \cong \mathbb{A}_{u,r}^2$ and its finite birational map to $\Sigma = V(\Delta)$ is the normalization. The triple-root locus $3ru = 2$ maps to Γ . Standard cubic-discriminant geometry gives an ordinary transverse cusp along Γ .

Proposition 4.1. *The full nonproperness set is $S_F = \Sigma$.*

Proof. Approaching D from $I \setminus D$ produces escaping source sequences with limits in $\pi(D) = \Sigma$. Conversely, properness of the finite map $I \rightarrow \mathbb{A}^3$ excludes any other finite asymptotic value. $\square$

Projection $I \rightarrow \mathbb{P}^1$ is a nontrivial affine-plane torsor under

$$E = \ker \left(\mathcal{O}^{\oplus 3} \xrightarrow{(2S^3, -S^2T, -T^3)} \mathcal{O}(3) \right) \cong \mathcal{O}(-1) \oplus \mathcal{O}(-2).$$

The displayed splitting follows from syzygies $(T, 2S, 0)$ and $(0, T^2, -S^2)$. There is no global section because constants p, q, r cannot produce the missing ST^2 term. Hence $\text{Pic}(I) \cong \mathbb{Z}$. Adjunction and the ramification formula give $K_I \cong \mathcal{O}(1) \cong \mathcal{O}(D)$, so D generates the divisor class group. Thus I is smooth affine but not factorial, whereas $I \setminus D \cong \mathbb{A}^3$ is factorial.

The existence of some finite completion is classical: Zariski's Main Theorem factors every separated quasi-finite morphism as an open immersion followed by a finite morphism [17]. The special content here is the explicit boundary, ramification, normalization, and torsor geometry, not the abstract existence of a completion.

5 Direct Dixmier and Poisson counterexamples

Let $J = DF$ and $B = J^{-1} = -\frac{1}{2} \text{adj}(J)$. Define

$$\delta_i = \sum_j B_{ji} \frac{\partial}{\partial x_j}.$$

Then $\delta_i(F_k) = \delta_{ik}$. The commutator $[\delta_i, \delta_j]$ annihilates $\mathbb{C}(F_1, F_2, F_3)$ and hence its finite separable extension $\mathbb{C}(x_1, x_2, x_3)$, so the vector fields commute.

For the third Weyl algebra, the assignments

$$X_i \mapsto F_i(X), \quad D_i \mapsto \sum_j B_{ji}(X) D_j$$

preserve all Weyl relations. They do not define an automorphism. Otherwise an automorphism would take the centralizer $\mathbb{C}[X]$ of $\mathbb{C}[X]$ onto the centralizer of $\mathbb{C}[F]$. Since every polynomial in X already centralizes $\mathbb{C}[F]$, this would force $\mathbb{C}[X] \subseteq \mathbb{C}[F]$, and hence equality, contradicting Theorem 1.1. Thus the Dixmier Conjecture fails in rank three.

On the canonical polynomial Poisson algebra, the assignments

$$x_i \mapsto F_i(x), \quad \xi_i \mapsto \sum_j B_{ji}(x) \xi_j$$

preserve brackets. The same centralizer argument proves noninvertibility, so the Poisson Conjecture fails in rank three. These implications are consistent with the classical equivalence framework [8].

5.1 A commuting volume-preserving frame and its incompleteness

The columns of B satisfy the additional exact identities

$$[\delta_i, \delta_j] = 0, \quad \text{div}(\delta_i) = 0, \quad \det(B) = -\frac{1}{2}.$$

The divergence equations are the Piola identity for the adjugate of a gradient matrix and are checked directly by the verifier. Thus the δ_i form an everywhere independent commuting, standard-volume-preserving polynomial frame dual to dP, dQ, dR .

Proposition 5.1 (Completeness criterion). *For a complex Keller map $G : \mathbb{A}^n \rightarrow \mathbb{A}^n$, let ∂_i be the commuting polynomial fields dual to dG_i . Then G is a polynomial automorphism if and only if all ∂_i are complete holomorphic vector fields. In particular, local nilpotence of all the ∂_i implies invertibility.*

Proof. For an automorphism the fields are pullbacks of constant coordinate fields. Conversely, their complete commuting flows give a $\mathbb{C}^n$ -action satisfying

$$G(\exp(s_1 \partial_1) \cdots \exp(s_n \partial_n) x) = G(x) + s.$$

The frame condition makes every orbit open, so connectedness gives one orbit; the formula makes the action free and G bijective. Ax–Grothendieck makes the inverse polynomial. $\square$

Theorem 5.2 (Total directional incompleteness). *Every nonzero constant linear combination $\delta_a = a_1 \delta_1 + a_2 \delta_2 + a_3 \delta_3$ is holomorphically incomplete and is not locally nilpotent.*

Proof. A local flow satisfies $F(\phi_t(x)) = F(x) + ta$. The omitted curve

$$\Gamma = \left\{ \left(q^2/12, q, 4/(3q) \right) : q \in \mathbb{C}^* \right\}$$

contains no affine line. Choose $\gamma \in \Gamma$ and $t_0 \neq 0$ such that $y = \gamma - t_0 a \notin \Gamma$. Since $F(\mathbb{A}^3) = \mathbb{A}^3 \setminus \Gamma$, choose x with $F(x) = y$. A complete flow would satisfy $F(\phi_{t_0}(x)) = \gamma$, a contradiction. $\square$

An explicit integral curve of δ_2 is

$$\gamma(t) = \left(2/t, -t/2, 5t^2/4 \right), \quad F(\gamma(t)) = (0, t, 0), \quad \gamma'(t) = \delta_2(\gamma(t)).$$

It escapes at the finite time $t = 0$. Thus the exact coframe defines a flat algebraic affine structure whose developing map is F , but every nonzero constant direction in its dual frame is incomplete.

6 Real fibers, a plane slice, and research directions

For real targets, the sign of Δ gives three real preimages when $\Delta > 0$, one when $\Delta < 0$, one on $\Sigma \setminus \Gamma$, and none on $\Gamma(\mathbb{R})$. Hence $F(\mathbb{R}^3) = \mathbb{R}^3 \setminus \Gamma(\mathbb{R})$.

If $E \subset \{\Delta > 0\}$ is relatively compact and measurable, the three real inverse branches and $|\det DF| = 2$ give

$$\text{vol}(F^{-1}(E)) = \frac{3}{2} \text{vol}(E).$$

For relatively compact measurable $E \subset \{\Delta < 0\}$, one real branch gives

$$\text{vol}(F^{-1}(E)) = \frac{1}{2} \text{vol}(E).$$

Every local orientation contribution is -1 , while the signed sum over a regular real fiber changes from -3 to -1 when two branches escape at the discriminant. This is not a global topological degree, since F is not proper.

Fix $P = p \neq 0$. The level surface is $X = \mathbb{A}^2 \setminus V(1 + xy)$, and the remaining coordinates are

$$Q_p = \frac{3px + xy^2 + y}{1 + xy}, \quad R_p = \frac{x(-px^2 + x^2y^2 + 3xy + 2)}{(1 + xy)^3},$$

with relative Jacobian $-2/(1+xy)^3$. This is an etale map from a punctured affine plane, not a counterexample on $\mathbb{A}^2$; the nonconstant unit $1+xy$ shows $X \not\cong \mathbb{A}^2$.

There is a second nonproperness diagnostic. Put

$$A = 1 + xy, \quad B = A^2z + y^2(4 + 3xy), \quad P = AB.$$

Since DF is invertible, dP never vanishes and $P : \mathbb{A}^3 \rightarrow \mathbb{A}^1$ is smooth. Nevertheless,

$$P^{-1}(p) \cong X \quad (p \neq 0), \quad P^{-1}(0) = \{B = 0\} \sqcup \{A = 0\} \cong X \sqcup (\mathbb{G}_m \times \mathbb{A}^1).$$

The zero-fiber components are disjoint because $A = 0$ forces $B = y^2 \neq 0$. Thus a smooth polynomial family has connected nonzero fibers but a disconnected zero fiber; the extra component enters from infinity.

6.1 Exact finite-field fiber statistics

Over every finite field $\mathbb{F}_q$ of odd characteristic, let $N_j(q)$ be the number of targets having exactly j rational preimages. Only $j = 0, 1, 3$ occur.

Theorem 6.1. *If $\text{char } \mathbb{F}_q \neq 2, 3$, then*

$$N_3 = \frac{(q-1)(q^2+2)}{6}, \quad N_1 = \frac{q^3+q^2-2q+2}{2}, \quad N_0 = \frac{(q-1)(q^2+2)}{3}.$$

In characteristic three,

$$N_3 = \frac{q^2(q-1)}{6}, \quad N_1 = \frac{q^2(q+1)}{2}, \quad N_0 = \frac{q^2(q-1)}{3}.$$

The limiting proportions for three, one, and zero preimages are $1/6, 1/2, 1/3$, the proportions of the three cycle types in S_3 .

Proof. A target determines the binary cubic $2pS^3 - qS^2T + 2ST^2 - rT^3$. Three rational preimages correspond to a three-element root subset of $\mathbb{P}^1(\mathbb{F}_q)$ whose ST^2 coefficient is nonzero, since that coefficient must be scaled to 2. Of the $\binom{q+1}{3}$ root triples, $(q-1)/2$ bad triples contain infinity and finite roots $a, -a$. For three finite roots, the bad equation is $ab + ac + bc = 0$; inversion turns it into three distinct nonzero elements with sum zero. Ordered-pair counting gives $(q-1)(q-5)/6$ bad finite triples when the characteristic is not three, and $(q-1)(q-3)/6$ in characteristic three. Subtraction gives N_3 . Since both source and target have q^3 points,

$$N_0 + N_1 + N_3 = q^3, \quad N_1 + 3N_3 = q^3,$$

which gives $N_0 = 2N_3$ and the stated formulas. □

The immediate research program is now sharply delimited:

1. Determine whether total-degree 4, 5, or 6 counterexamples exist in dimension three. Degree at most three is known [6, 7].
2. Produce and optimize explicit cubic-homogeneous and Druzkowski forms via the genuine stable reductions [2]; a naive circuit lift is invalid.
3. Classify the affine-plane torsors and ramification complements producing $\mathbb{A}^3$.

4. Classify weighted-lift families and their possible monodromy groups. A same-day note constructs every generic degree at least three [14]; we do not claim priority for that family.
5. Seek explicit small witnesses for related failed conjectures. Published implications give failure of the Mathieu conjecture for $SU(3)$ and existential failure in some finite dimension of the Gaussian Moments, Vanishing, and Image conjectures [13, 12, 10, 9, 11].
6. Study whether the deleted-hyperbola slice yields an obstruction or a useful deformation for the still-separate plane problem.
7. Classify the pole orders, escape divisors, and degree growth of the incomplete inverse-Jacobian flows.
8. Use the weighted $\mathbb{C}^*$ -action to form quotient surfaces and classify equivariant Keller maps.
9. Formalize the determinant, collision, and inverse cubic in a proof assistant.
10. Extend Theorem 6.1 to extension-field zeta functions, arithmetic monodromy, higher-degree lift families, and bad reduction.

An earlier informal assertion that the map was “forced in a natural minimal PDE ansatz” is deliberately omitted: the class and equivalence relation were not specified precisely enough for a publication-grade minimality theorem.

7 Reproducibility and limitations

The repository verifies the determinant, collision, weighted equivariance, collision family, weighted quotient, the quotient square-law differential identities, inverse identities, discriminants, boundary parametrization, nonproper path, slice Jacobian, first-coordinate factorization, the explicit escaping integral curve, finite-field regression counts, and commuting divergence-free inverse-Jacobian fields using exact symbolic arithmetic. These checks certify polynomial identities, not literature priority or community acceptance. This note is not peer reviewed, not an arXiv submission, and not a substitute for independent expert review. Its status discussion is time-stamped and may be superseded.

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